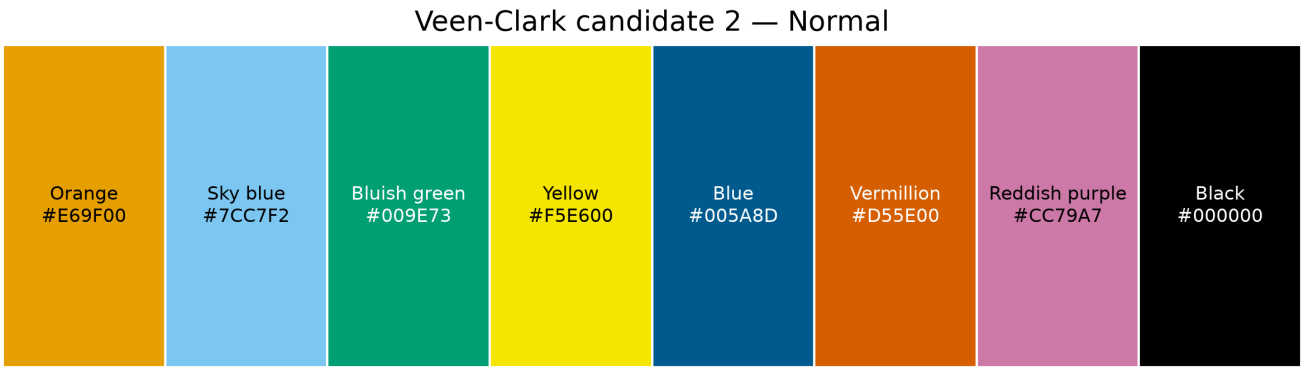
Veen-Clark palette

Candidate 2 evaluation



Candidate palette in normal sRGB.

An experimental Okabe-Ito derivative by Roland J. Veen and Charlie Clark. Sky blue is lightened and blue is darkened so the pair can represent related subgroups in scientific scatter and bar charts. Candidate 2 copies candidate 1 with yellow changed from #F0E442 to #F5E600, aiming for a more distinct, less green, warmer-looking yellow in presentations; all other colours are unchanged.

Status: candidate. This report compares colour-difference metrics for the candidate and Okabe-Ito palettes.

Generated with PaletteBench 0.1.0 using Python 3.11.16. Exact inputs, hashes, dependencies, full-precision CSV/JSON data, and individual vector figures accompany this PDF in the repository.

Results

The intended sky-blue/blue subgroup pair has greater modelled separation relative to Okabe-Ito under every standard reported condition: normal, protan 100%, deutan 100%, tritan 100%, and grayscale.

Condition	Okabe-Ito	Candidate	Change
Normal	22.31	37.92	+15.61
Protan 100%	20.75	36.03	+15.28
Deutan 100%	22.45	38.57	+16.12
Tritan 100%	19.95	35.08	+15.13
Grayscale	21.62	37.43	+15.81

The whole-palette result is mixed. The normal-vision 10th-percentile change is +2.73 Delta E00 and the protan-100% minimum change is +3.56. Darkening blue reduces separation from black and produces the largest loss against vermillion under tritan 100% (-12.20 Delta E00).

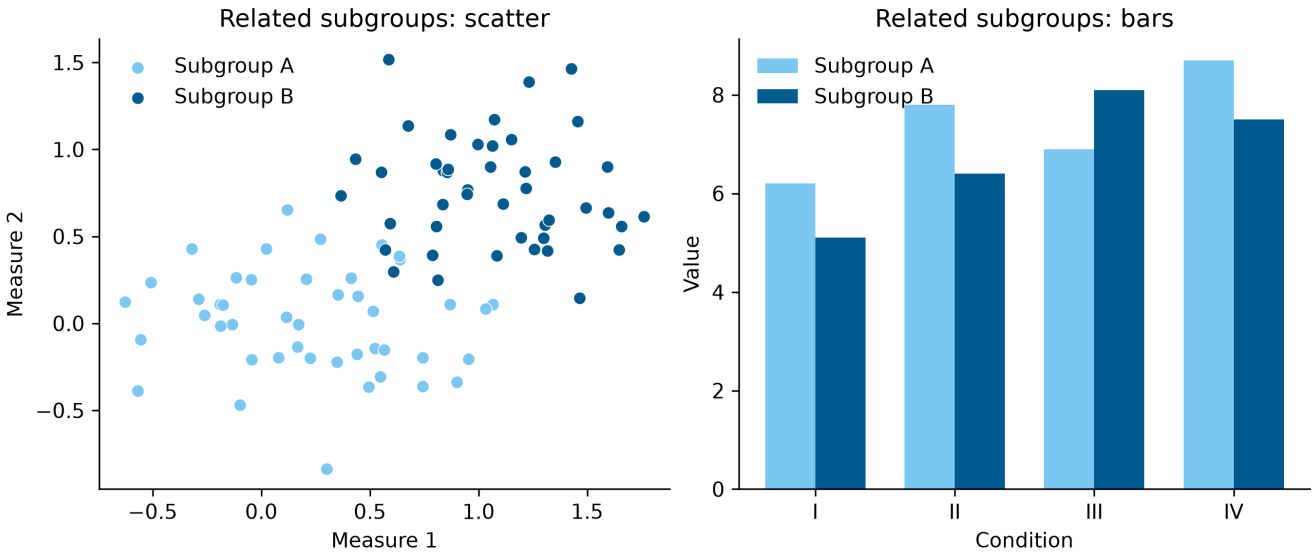


Figure 1. Synthetic data illustrating the two blues with circular markers and solid bar fills.

Yellow change: rationale and evidence

Candidate 2 retains candidate 1's lighter sky blue and darker blue. Its additional change replaces the original Okabe-Ito yellow #F0E442 with #F5E600. The design aim is a more distinct, less green, warmer-looking yellow that stands out more clearly in presentation figures.

This choice was motivated by informal feedback from people with regular colour vision: in presentations with low colour accuracy, the original yellow was reported to read too much like green and not stand out enough. This feedback provides the design rationale; it was not a controlled participant study. The intended warmer, less green appearance is a perceptual design description, not a measured improvement across presentation systems.

Yellow-specific modelled separation

The table isolates yellow's relationship with orange and bluish green. These two colours are unchanged from Okabe-Ito and candidate 1, so these differences are attributable to the yellow replacement. Values are CIEDE2000 (Delta E00); positive changes indicate increased modelled separation.

Condition	Yellow paired with	Original	Candidate 2	Change
normal	orange	21.73	22.04	+0.31
normal	bluish green	38.04	39.94	+1.90
protan100	orange	15.17	15.81	+0.64
protan100	bluish green	28.05	28.93	+0.89
deutan100	orange	11.61	12.30	+0.70
deutan100	bluish green	33.90	35.47	+1.58
tritan100	orange	18.83	19.15	+0.32
tritan100	bluish green	43.57	43.25	-0.33
grayscale	orange	12.83	13.29	+0.47
grayscale	bluish green	23.32	23.78	+0.46

These results describe ideal sRGB and the specified CVD/grayscale simulations. They do not test projector or display colour errors, ambient light, or audience perception. In particular, greater pairwise separation does not establish that yellow will stand out more on a white slide background. Confirming the presentation benefit requires viewing comparisons under the intended conditions.

Baseline-relative comparison

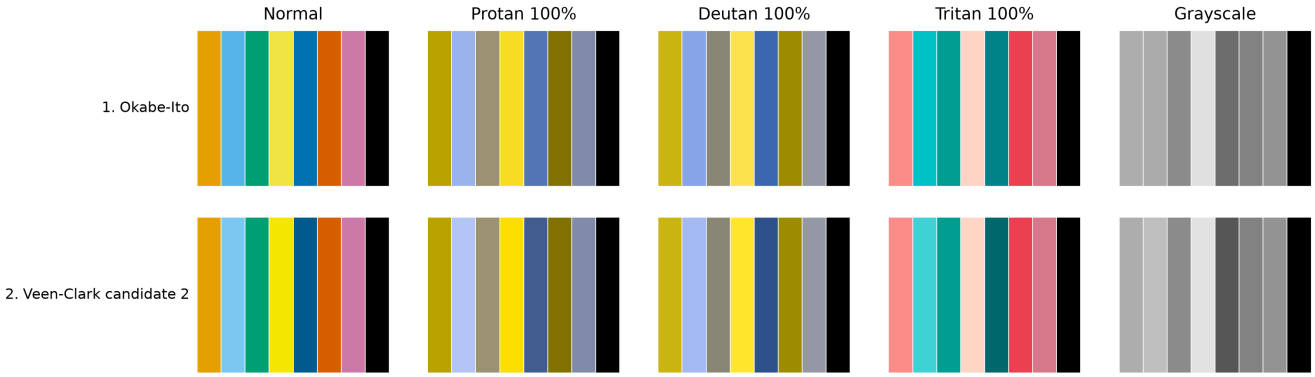


Figure 2. Okabe-Ito baseline and Veen-Clark candidate under the standard reported conditions.

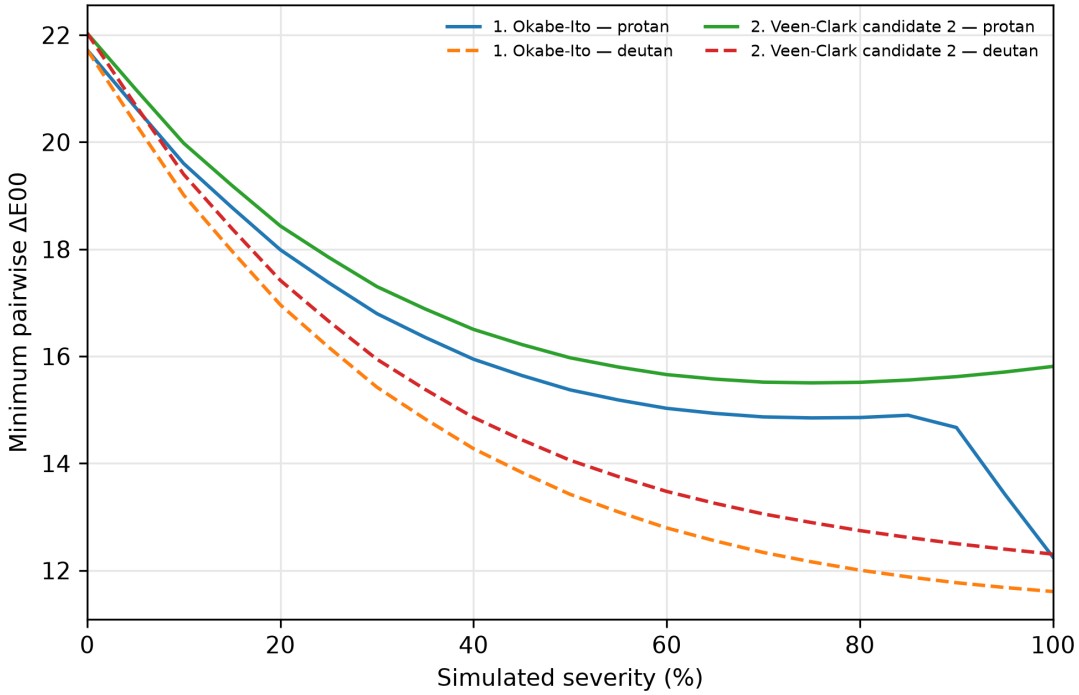


Figure 3. Whole-palette minimum Delta E00 by condition.

Baseline-relative comparison

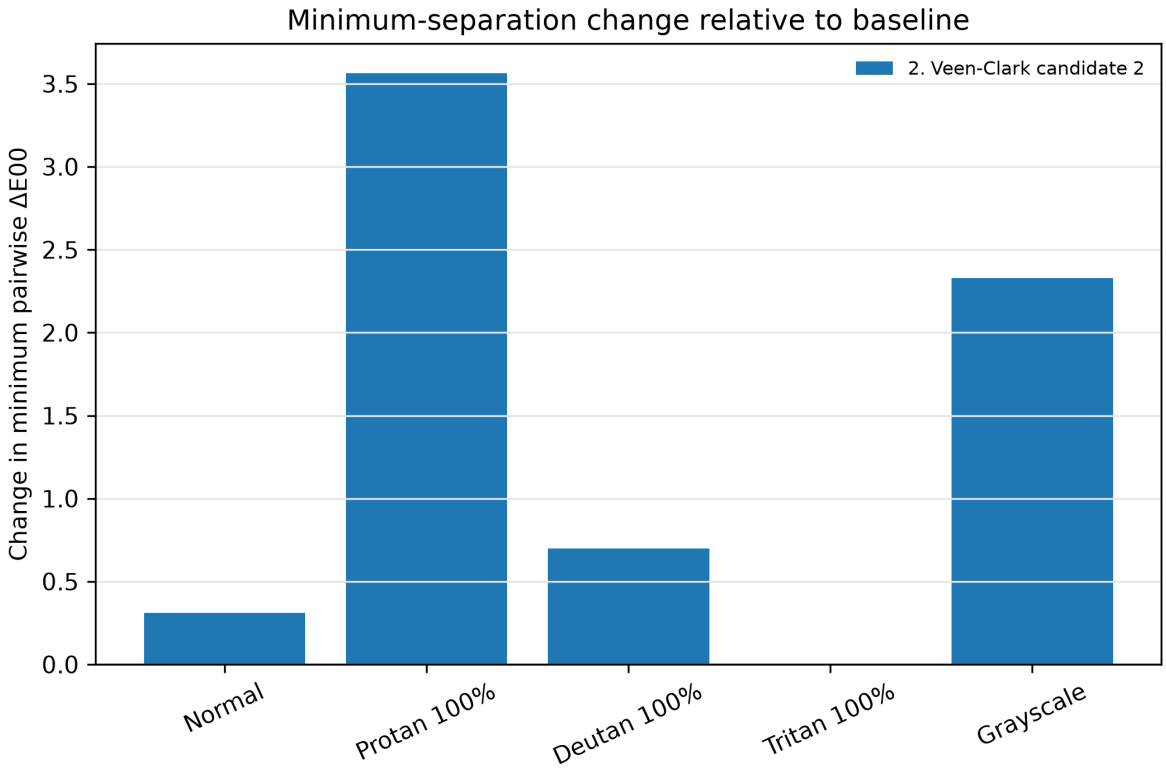


Figure 4. Change in minimum separation relative to Okabe-Ito.

Positive changes indicate greater modelled separation; negative changes indicate reduced modelled separation. Zero change means the minimum separation is unchanged.

Methods and assumptions

Six-digit values are interpreted as gamma-encoded sRGB. PaletteBench applies Colorspacious sRGB1+CVD simulations based on Machado, Oliveira and Fernandes, clips results to displayable sRGB, converts through CIE XYZ to CIE Lab under the sRGB D65 white point, and calculates CIEDE2000 colour differences with Colour Science for Python. Grayscale preserves WCAG relative luminance.

Severity percentages are simulation parameters, not clinical measurements. Partial tritan severity is omitted from the standard report. Clipping provenance and raw values are retained in the machine-readable audit directory.

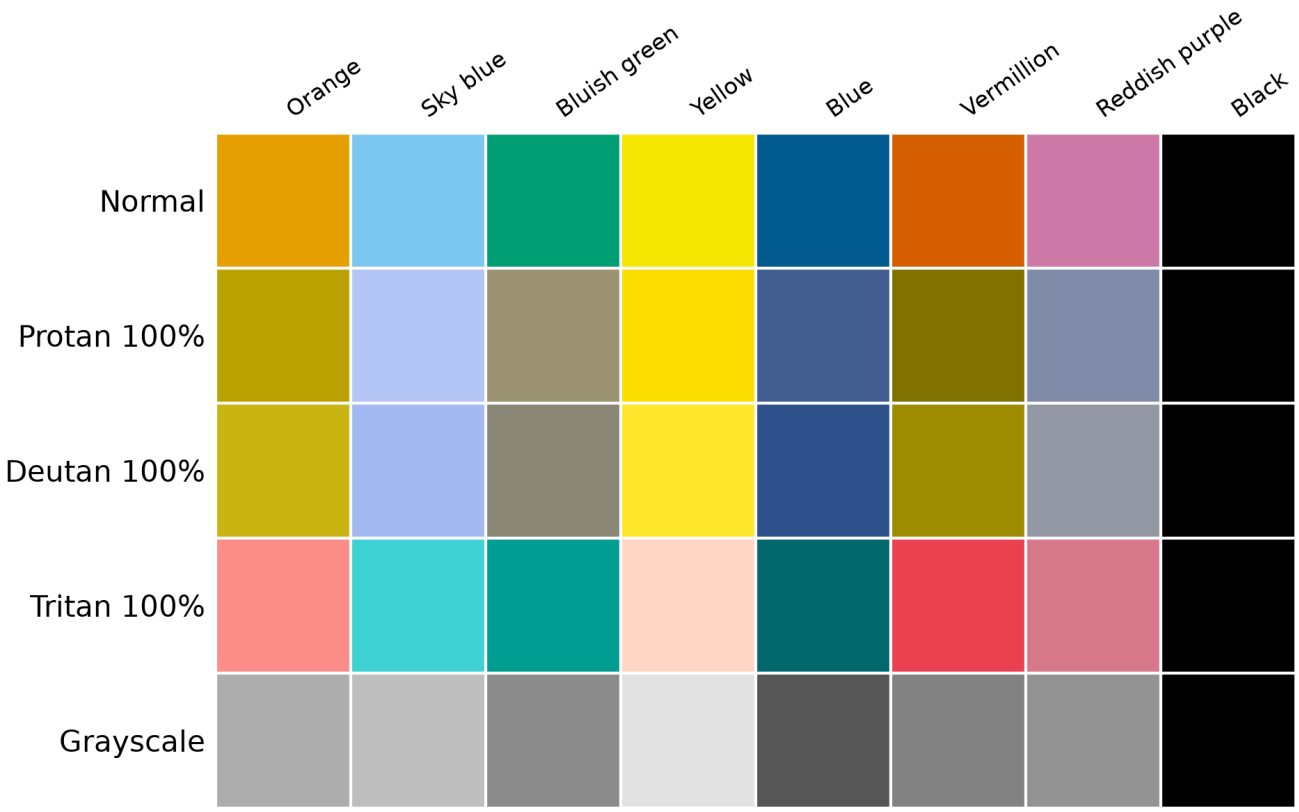


Figure 5. Normal, simulated CVD, and grayscale overview.

Palette characteristics

Colour	Hex	RGB	L*	Chroma	Luminance	Group
Orange	#E69F00	230 159 0	70.61	76.48	0.4162	-
Sky blue	#7CC7F2	124 199 242	77.01	30.81	0.5154	blue_family
Bluish green	#009E73	0 158 115	57.74	47.49	0.2569	-
Yellow	#F5E600	245 230 0	89.86	89.87	0.7601	-
Blue	#005A8D	0 90 141	36.43	34.62	0.0924	blue_family
Vermillion	#D55E00	213 94 0	54.19	76.67	0.2215	-
Reddish purple	#CC79A7	204 121 167	61.05	39.97	0.2930	-
Black	#000000	0 0 0	0.00	0.00	0.0000	-

Condition-level summary

Condition	Min	Weakest pair	Mean	Median	10th pct.
normal	22.04	orange / yellow	49.68	49.65	33.71
protan20	18.43	orange / yellow	46.97	47.83	33.91
protan40	16.50	orange / yellow	45.05	44.26	28.67
protan60	15.66	orange / yellow	43.14	42.95	23.52
protan80	15.51	orange / yellow	41.88	42.61	19.46
protan100	15.81	orange / yellow	41.32	41.46	18.08
deutan20	17.42	orange / yellow	46.89	47.93	33.41
deutan40	14.86	orange / yellow	45.12	42.83	29.74
deutan60	13.48	orange / yellow	43.37	40.27	23.95
deutan80	12.74	orange / yellow	42.23	38.78	20.38
deutan100	12.30	orange / yellow	41.70	37.53	16.01
tritan100	10.87	orange / reddish_purple	44.53	52.05	15.79
grayscale	2.93	bluish_green / reddish_purple	26.21	20.88	5.79

Palette strips

Veen-Clark candidate 2 — Normal



Normal sRGB

Veen-Clark candidate 2 — Protan 100%



Simulated protan 100%

Veen-Clark candidate 2 — Deutan 100%



Simulated deutan 100%

Veen-Clark candidate 2 — Tritan 100%



Simulated tritan 100%

Veen-Clark candidate 2 — Grayscale



WCAG-luminance grayscale

Severity progressions

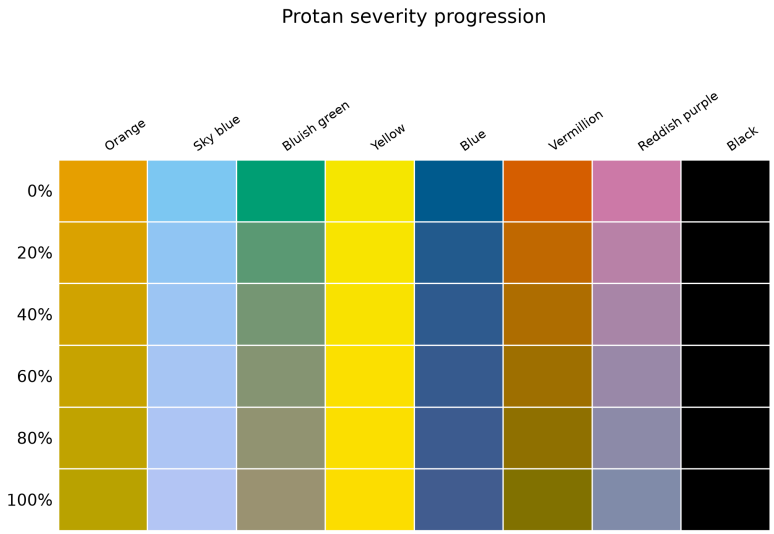


Figure 6. Protan simulation progression from 20% to 100%.

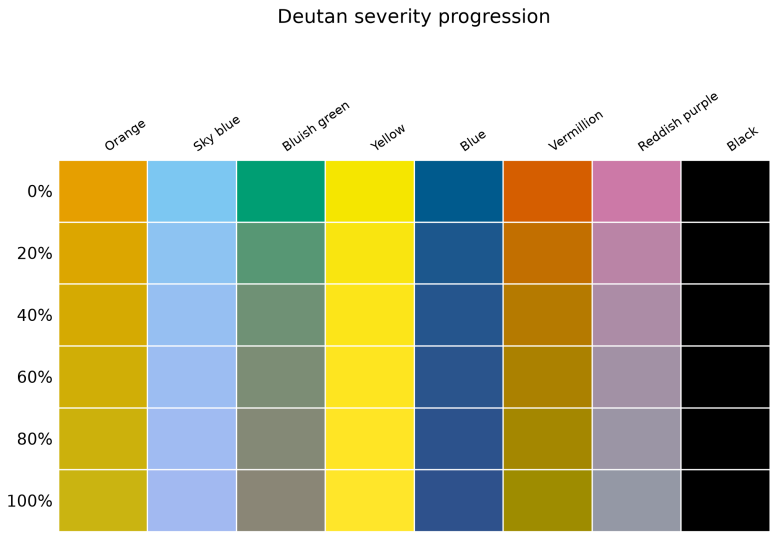


Figure 7. Deutan simulation progression from 20% to 100%.

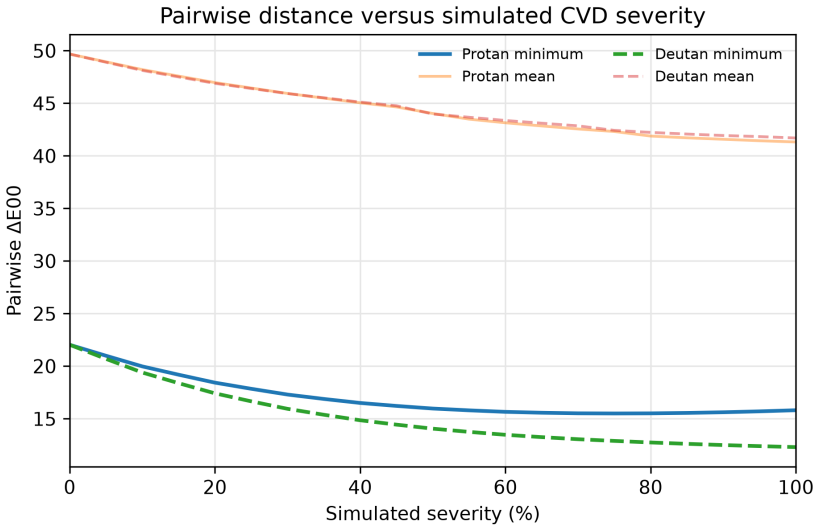
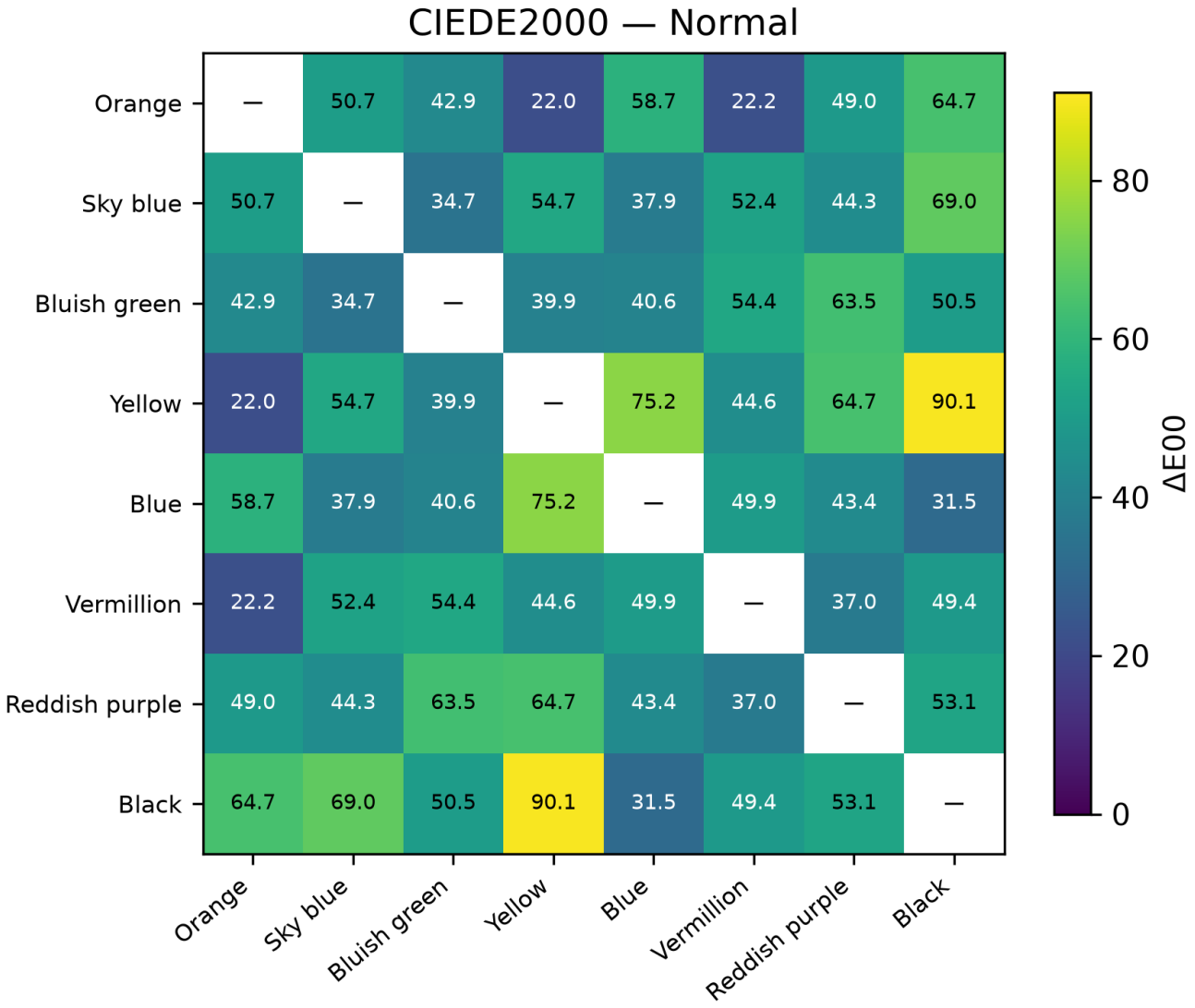


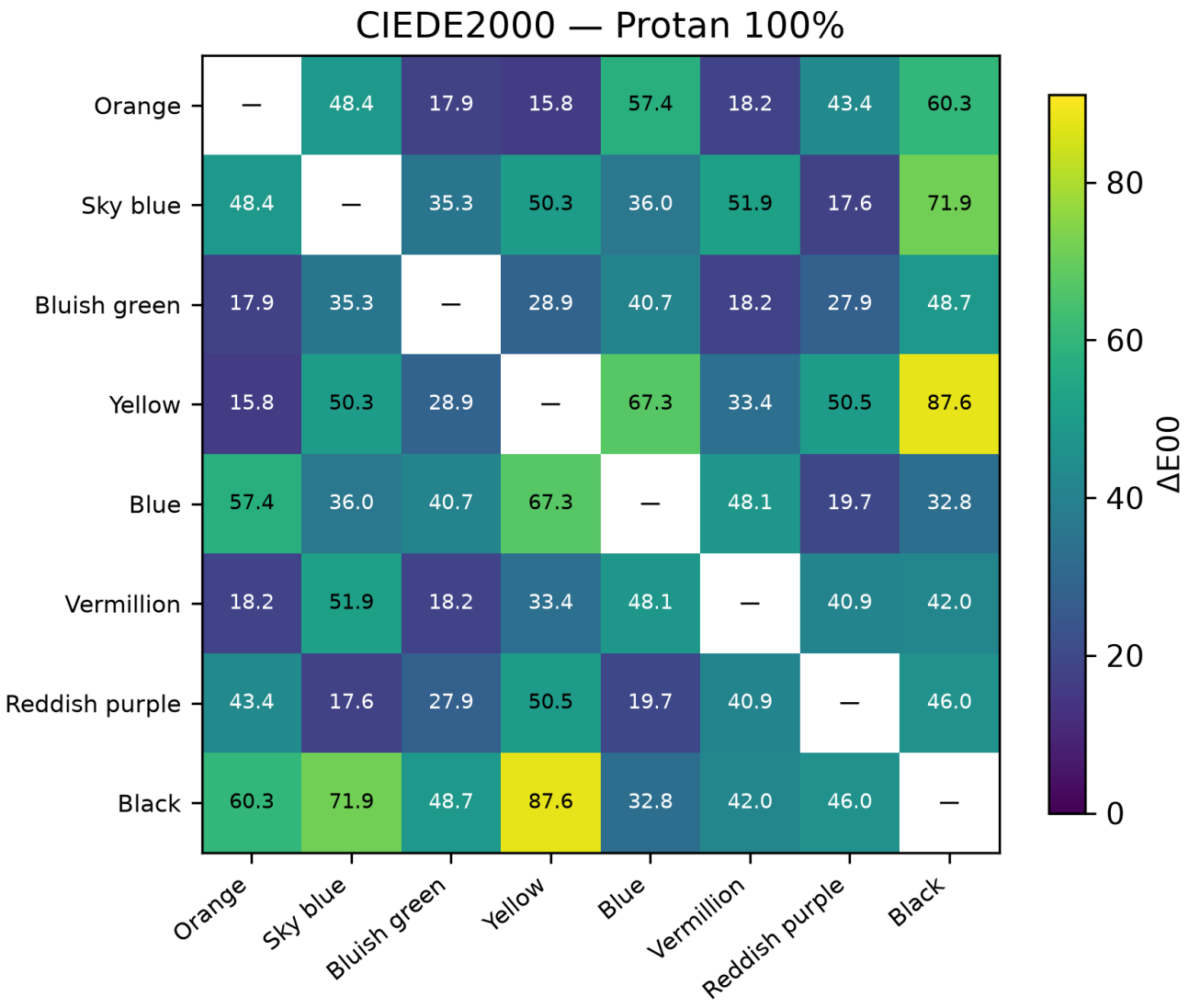
Figure 8. Minimum pairwise distance across simulated severity.

Normal pairwise heatmap



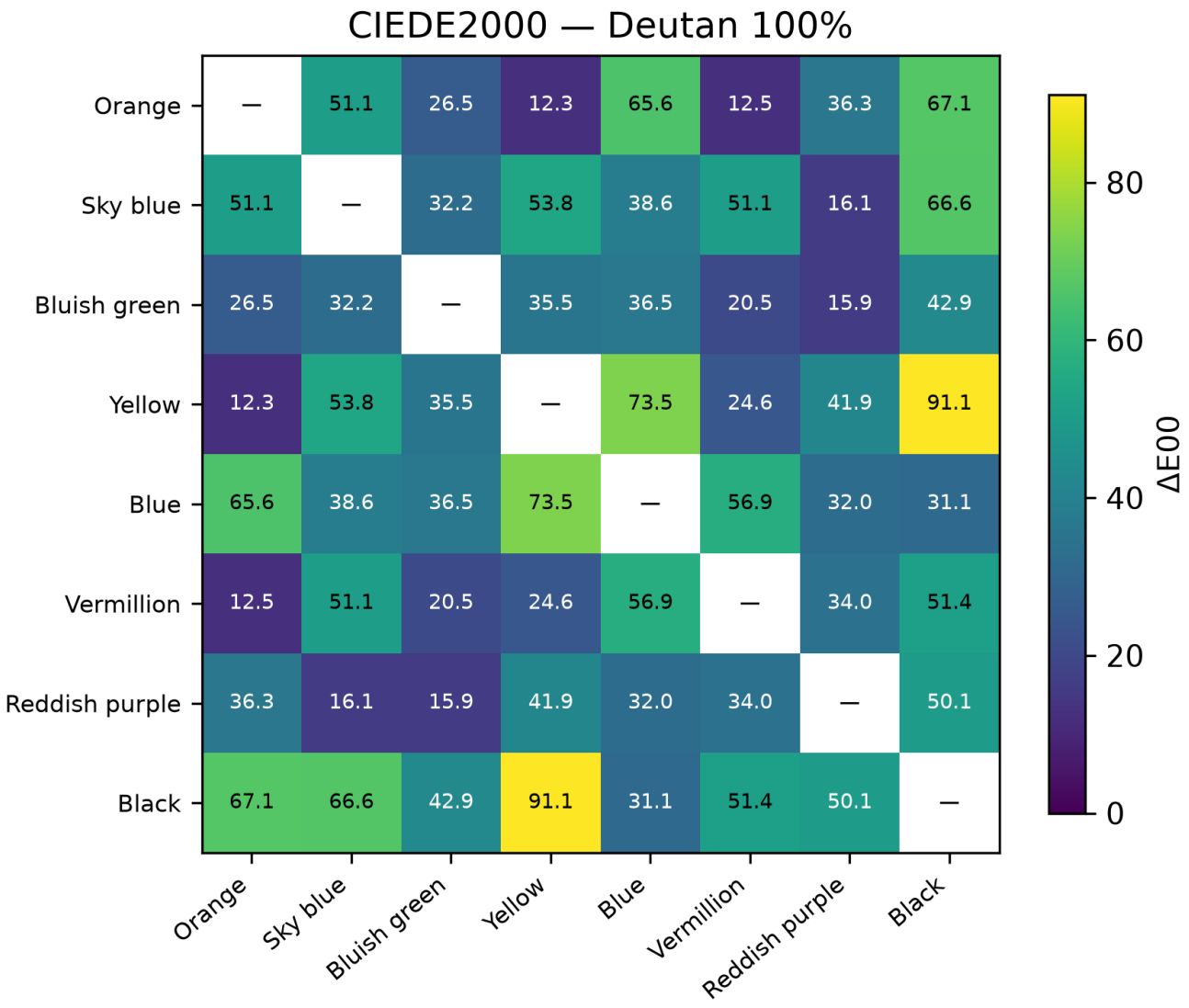
Normal pairwise heatmap.

Protan 100% pairwise heatmap



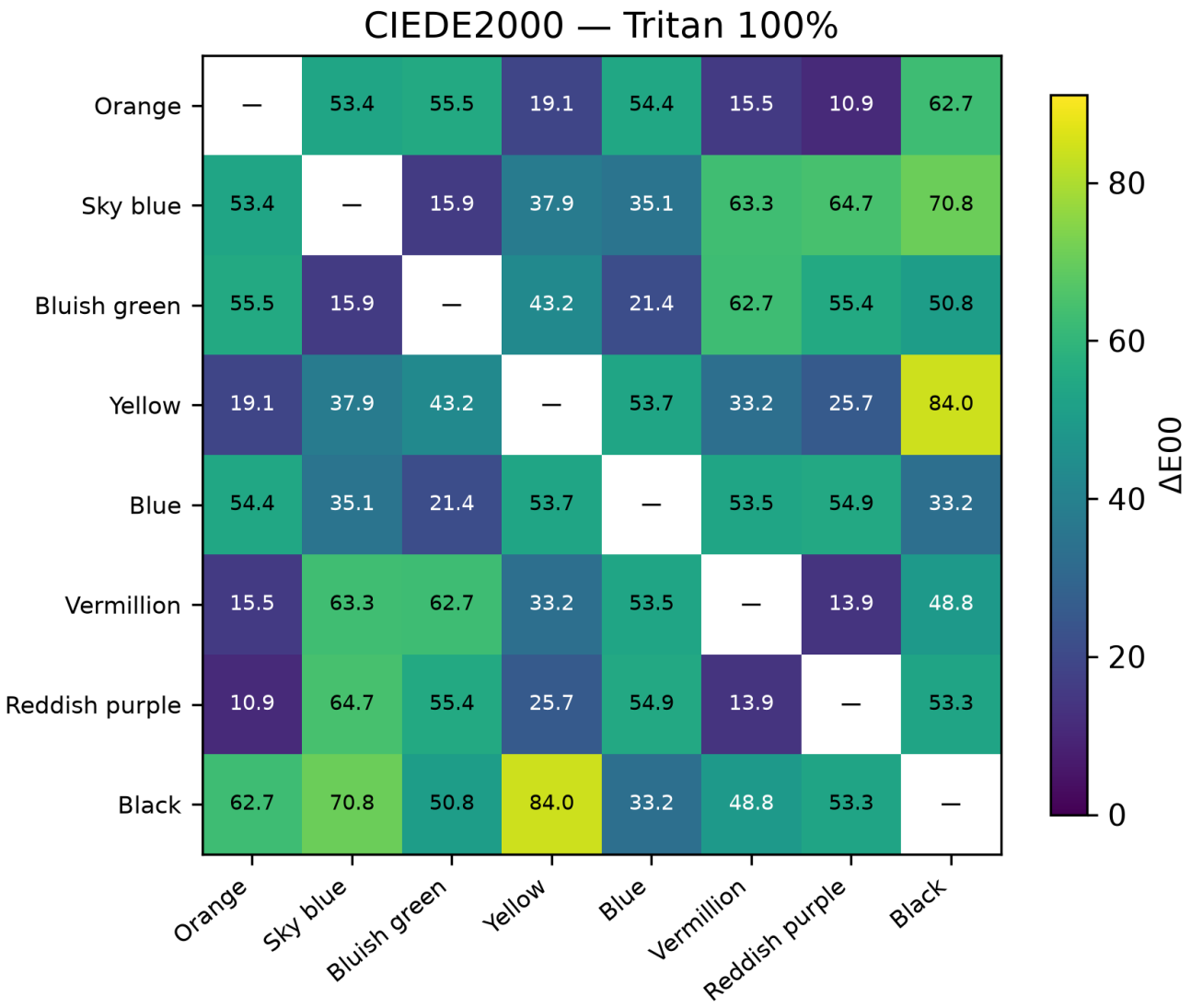
Protan 100% pairwise heatmap.

Deutan 100% pairwise heatmap



Deutan 100% pairwise heatmap.

Tritan 100% pairwise heatmap



Tritan 100% pairwise heatmap.

Normal diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Normal diagonal split pair matrix.

Protan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Protan 100% diagonal split pair matrix.

Deutan 100% diagonal split pair matrix

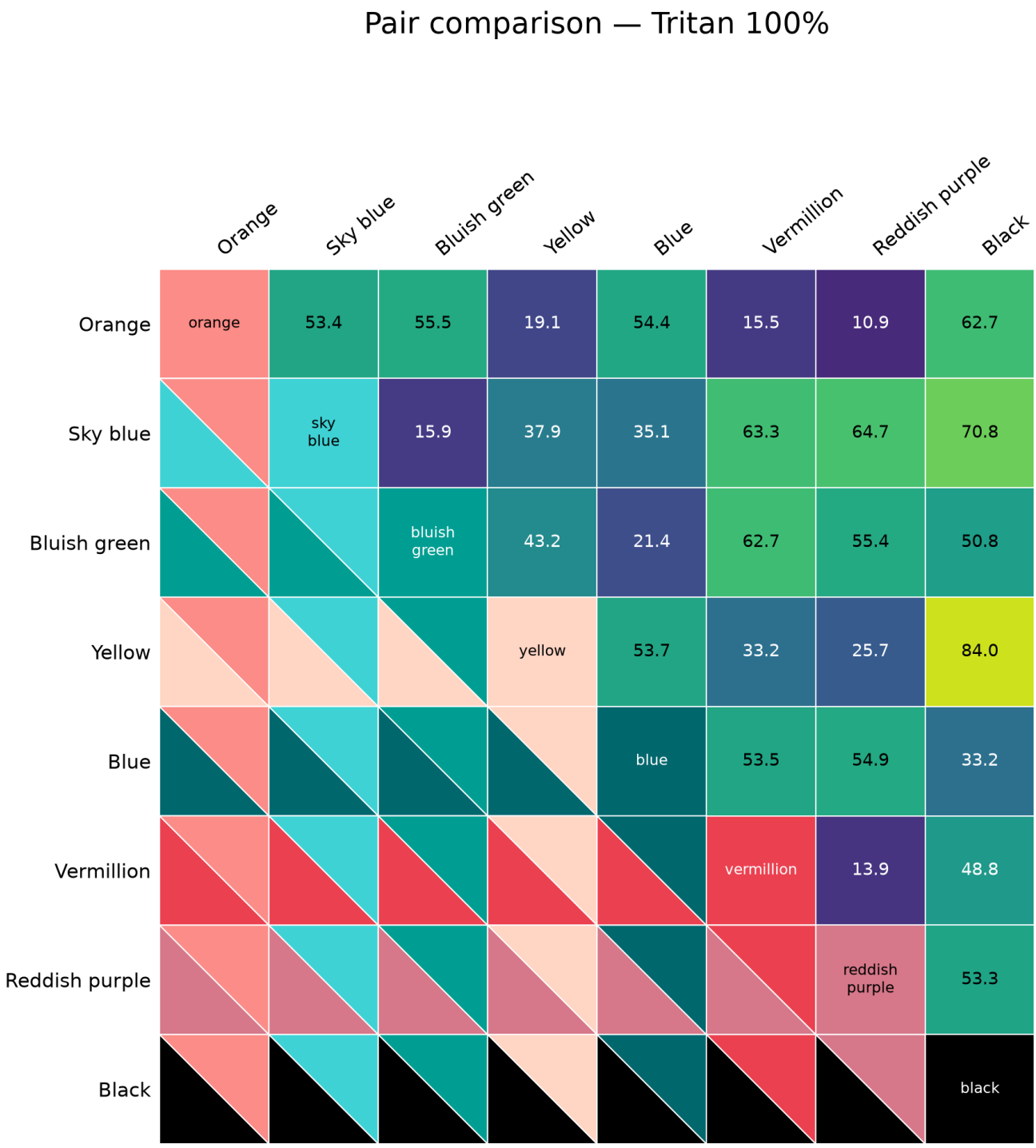
The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Deutan 100% diagonal split pair matrix.

Tritan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Tritan 100% diagonal split pair matrix.

Grouped blue-family analysis

The YAML declares sky blue and blue as one visual family. With two members, the group contains one colour pair: sky blue and blue.

Condition	Severity	Pairs	Minimum	Mean	Maximum
normal	0	1	37.92	37.92	37.92
protan20	20	1	37.32	37.32	37.32
protan40	40	1	36.87	36.87	36.87
protan60	60	1	36.53	36.53	36.53
protan80	80	1	36.26	36.26	36.26
protan100	100	1	36.03	36.03	36.03
deutan20	20	1	37.83	37.83	37.83
deutan40	40	1	37.92	37.92	37.92
deutan60	60	1	38.10	38.10	38.10
deutan80	80	1	38.32	38.32	38.32
deutan100	100	1	38.57	38.57	38.57
tritan100	100	1	35.08	35.08	35.08
grayscale	0	1	37.43	37.43	37.43

Weakest-pair diagnostic



Figure 9. Weakest pairs under each standard condition.

Interpretation and limitations

The candidate improves its intended blue-family separation in all standard PaletteBench conditions. It does not improve every pair. In particular, the darker blue approaches black, and its tritan-simulated relationship with vermillion loses separation relative to Okabe-Ito. Redundant encodings remain appropriate whenever category identity is critical.

CVD simulations and perceptual-distance models do not reproduce a particular person's perception. Monitor calibration, viewing environment, acuity, ageing, print reproduction, surrounding colours, spatial context, and mark size can all change discriminability. The yellow revision responds to informal regular-vision feedback about presentations. Its intended warmer, less green and more distinct appearance has not been established in a controlled viewing study. Formal human evaluation and presentation-device validation are outside this candidate audit.

Reproducibility

Palette input SHA-256: 32c8b4b23a6b8b345afb9b362a1543a8e00c76c89c8e78121baf3eccec94f696. PaletteBench version: 0.1.0. Metric: CIEDE2000 via colour-science; sRGB D65 to CIE Lab. Simulation: Colorspacious sRGB1+CVD (Machado et al.) with display-bound clipping.

References

Okabe, M. and Ito, K. (2002). Color Universal Design: How to make figures and presentations that are friendly to colorblind people.

Wong, B. (2011). Points of view: Color blindness. Nature Methods 8, 441. doi:10.1038/nmeth.1618; corrected 2023, doi:10.1038/s41592-023-01974-0.

Veen, R. J. (2026). PaletteBench 0.1.0. Zenodo. doi:10.5281/zenodo.23071831.